

Simple Discretization: Binning

❑ **Equal-width (distance) partitioning**

- ❑ Divides the range into N intervals of equal size: uniform grid
- ❑ if A and B are the lowest and highest values of the attribute, the width of intervals will be: $W = (B - A)/N$.
- ❑ The most straightforward, but outliers may dominate presentation
- ❑ Skewed data is not handled well

❑ **Equal-depth (frequency) partitioning**

- ❑ Divides the range into N intervals, each containing approximately same number of samples
- ❑ Good data scaling
- ❑ Managing categorical attributes can be tricky

Example: Binning Methods for Data Smoothing

❑ Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

* Partition into equal-frequency (**equal-depth**) bins:

- Bin 1: 4, 8, 9, 15
- Bin 2: 21, 21, 24, 25
- Bin 3: 26, 28, 29, 34

* Smoothing by **bin means**:

- Bin 1: 9, 9, 9, 9
- Bin 2: 23, 23, 23, 23
- Bin 3: 29, 29, 29, 29

* Smoothing by **bin boundaries**:

- Bin 1: 4, 4, 4, 15
- Bin 2: 21, 21, 25, 25
- Bin 3: 26, 26, 26, 34